

Kaveh Mousavand

Okinawa Institute of Science and Technology (OIST), Okinawa, Japan

☎ +81(098)982-3540 & +1(437)580-6262

✉ kaveh.mousavand@oist.jp & mousavand.kaveh@gmail.com

🌐 <https://kaveh-mousavand.github.io/index.html>

Employment

- | | |
|--|-----------------------|
| Okinawa Institute of Science and Technology (OIST) | Japan |
| * <i>Staff Scientist in Representation Theory & Algebraic Combinatorics</i> | (July 2025-Now) |
| Okinawa Institute of Science and Technology (OIST) | Japan |
| * <i>Postdoctoral Scholar in Representation Theory & Algebraic Combinatorics</i> | (July 2023-June 2025) |
| Queen's University | Canada |
| * <i>Coleman Postdoctoral Fellow in Department of Mathematics & Statistics</i> | (July 2020-June 2023) |

Research Interests

Representation Theory of Algebras, Combinatorics & Cluster Algebras, Topological Data Analysis.

Education

- | | |
|---|---------------|
| Doctoral studies at Université du Québec à Montréal | Canada |
| ○ <i>Ph.D. dissertation: τ-tilting finiteness of minimal representation-infinite algebras</i> | 2015-2020 |
| Ph.D. Advisor: Prof. Hugh Thomas | |
| <i>I started my PhD studies at University of New Brunswick and then moved to UQAM, after my advisor, Prof. Thomas, accepted a new position and moved to Montréal.</i> | |
| Master's program at Chamran University of Ahvaz | Iran |
| ○ <i>MSc. in Algebra, Supervisor: Prof. Omid Ali Karamzadeh</i> | 2009-2011 |
| Thesis: Subrings and Maximal Subrings of Zero-dimensional Rings. | |
| Undergraduate program at Yasouj University | Iran |
| ○ <i>BSc. in Pure Mathematics</i> | 2003-2007 |

Publications (in chronological order)

1. (with S. Asai, O. Iyama, C. Paquette), *Left modular and extremal lattices through brick-directed algebras*, Proc. of the 38th Conf. on Formal Power Series and Algebraic Combinatorics (2026).
2. (with C. Paquette), *Hom-Orthogonal modules and brick-Brauer-Thrall conjectures*, J. of Algebra, **686** (2026), 650-676.
3. (with M. Kulkarni, J. Matherne, D. J. Rock), *A continuous associahedron of type A*, Math. Z. **312**, 9 (2026).
4. (with C. Paquette), *Biserial algebras and generic bricks*, Math. Z., **310**, 64 (2025).
5. (with V. Bazier-Matte, N. Chapelier-Laget, G. Douville, H. Thomas, E. Yıldırım), *ABHY associahedra and Newton polytopes of F -polynomials for cluster algebras of simply laced finite type*, J. Lond. Math. Soc. (2), **109** (2024), no. 1.

6. τ -tilting finiteness of biserial algebras, *Algebr. Represent. Theory*, **26** (2023), no. 6, 2485–2522.
7. (with C. Paquette), *Minimal τ -tilting infinite algebras*, *Nagoya Math. J.* **249** (2023), 221–238.
8. τ -tilting finiteness of non-distributive algebras and their module varieties, *J. Algebra*, **608** (2022), 673–690.
9. (with T. Brüstle, G. Douville, H. Thomas, E. Yildirim), *On the combinatorics of gentle algebras*, *Canad. J. Math.* **72** (2020), no. 6, 1551–1580.
10. (with A. Garver, T. McConville), *A categorification of biclosed sets of strings*, *J. Algebra*, **546** (2020), 390–431.
11. (with D. Kucerovsky and A. Sarraf), *On some properties of Toeplitz matrices*, *Cogent Math.* **3** (2016), Art. ID 1154705.

Preprints (under review):

12. (with C. Paquette), *A Proof of the Second brick-Brauer-Thrall Conjecture for E-infinite algebras* (arXiv:2609.31417).
13. (with C. Paquette), *Brick infinite algebras admit infinitely many non- τ -rigid bricks* (arXiv:2606.27063).
14. (with S. Asai, O. Iyama, C. Paquette), *Left modularity and extremality for (some) infinite lattices* (arXiv:2604.20947).
15. (with M. Deaconu, C. Paquette), *Infinite string bricks and Sturmian words over some gentle algebras* (arXiv:2510.22377).
16. (with S. Asai, O. Iyama, C. Paquette), *Brick-splitting torsion pairs and left modular Lattices* (arXiv:2506.13602).
17. (with C. Paquette), *Geometric interactions between bricks and τ -rigidity* (arXiv:2311.14863).

Survey:

18. (with C. Paquette), *On the bricks (Schur representations) of finite dimensional algebras* (arXiv:2508.11789).

Applet:

19. *FDB Applet*: A non-profit self-contained applet for exploring finite-dimensional gentle algebras defined by quivers with relations (available on my website).

Repository:

20. *Brick Lodge*: A non-profit growing repository of important developments on some open conjectures, and a space for sharing new results, open problems, and exchanging ideas (available on my website).

Grants & Awards

- Fields Institute Elliott-Yui Distinguished Visitor (EYDV) Fellowship (2026).
- Tomlinson Fellowship, Visiting Scholar Funding Allocation, Bishop's University, Canada (2025-26).
- SHINKA Grant, Partnership with University of Tokyo: Prof. O. Iyama and Dr. S. Asai (2025).
- KAKENHI Early Career Research Grant (2024-27).
- Queen's University Professional Development Award (2022).
- Mitacs Globalink Research Award (2019).
- Scholarship for outstanding PhD candidates, Institute des Science du Mathématiques (2018).
- Bourse de stage international du FRQNT (2017).
- Bourse d'excellence, Institute des Science du Mathématiques (2016, 2017).

Research Visits (longer than 1 month)

- Queen's University, Kingston, Canada (2 months: Oct-Nov of 2026).
- Laboratoire d'Algèbre, de Combinatoire et d'Informatique Mathématique (LACIM), Montréal, Canada (2.5 months: March-May of 2026).
- Bishop's University, Sherbrooke, Canada (2 months: Sept-Oct of 2025).
- Bielefeld Representation Theory Group, Bielefeld, Germany (1.5 months: Oct-Nov of 2023).
- Institut Henri Poincaré (IHP), Paris, France (3 months: Jan-March of 2020).
- University of Paris-Sud, France (3 months: Sept-Nov of 2019).
- Massachusetts Institute of Technology (MIT), MA, US (4 months: Feb-May of 2017).
- Banach Center, Warsaw (IMPAN), Poland (3 months: Sept-Nov of 2016).

Selected presentations (Only at Conferences & Workshops after 2020)

1. "*Lattice of torsion classes and (generalized) left modularity and extremality*", International Conference on Representations of Algebras (ICRA 22), Grenoble, France (July 2026).
2. "*Directedness of indecomposables, bricks, and spread (interval) modules*", Topological Data Analysis and Representation Theory 2026, Tohoku University, Japan (February 2026)*[Invited Speaker].
3. "*Recent developments on brick-Brauer-Thrall conjectures*", Quivers in Representation Theory, Chennai Mathematical Institute, India (December 2025)*[Invited Speaker].
4. "*Brick-directed algebras and their applications*", Program on Combinatorics, Geometry, and Representation Theory, ICTS Bangalore, India (November 2025).
5. "*Minimal brick-infinite tame algebras*", 35th Meeting on the Rep. Theory of Algebras & Related Topics (Conference in honor of Thomas Brüstle), Sherbrooke, Canada (Oct 2025)*[Invited Speaker].
6. "*Endomorphism algebra of modules and their modern applications*", Representation Theory of Lie Algebras, Shenzhen International Center for Mathematics, China (May 2025)*[Invited Speaker].
7. "*Hom-orthogonal modules and brick-Brauer-Thrall conjectures*", Perspectives in Tilting Theory and Related Topics, RIMS, Kyoto, Japan (Feb 2025)*[Invited Speaker].
8. "*Hom-orthogonal modules, bricks and representation varieties*", Topological Data Analysis and Representation Theory, Kobe, Japan (Feb 2025)*[Invited Speaker].
9. "*Bricks and hom-orthogonal modules*", Autumn meeting of Mathematical Society of Japan, Osaka, Japan (Sept 2024).
10. "*brick-Brauer-Thrall conjectures and some applications*", International Conference on Representations of Algebras (ICRA 21), Shanghai, China (Aug 2024).
11. "*New realizations & generalizations of locally representation-directed algebras*", International Workshop on Homological Approximation Theory and Applications, Hangzhou, China (July 2024)*[Invited Speaker].
12. "*A (τ) -tilting counterpart of cluster algebras of minimal infinite type*", Advances in Cluster Algebras, Nagoya University, Japan (March 2024)*[Invited Speaker].
13. "*Distribution of bricks and one-parameter families of stable modules*", McKay correspondence, Tilting theory and related topics, Kavli IPMU, Japan (Dec 2023)*[Invited Speaker].
14. "*The 2nd brick-Brauer-Thrall conjecture & a torus action on varieties*", Tilting in Representation Theory, Singularities and Noncommutative Geometry, Oaxaca, Mexico (Sept 2023).
15. "*A combinatorial account of τ -tilting finiteness*", Spring School in Algebra and Representation Theory, University of Kent in Canterbury (April 2023).

16. “Algebraic and combinatorial treatments of g -vector fans”, Combinatorial Algebra meets Algebraic Combinatorics, Waterloo, Canada (Jan 2023).
17. “Biserial algebras and generic bricks”, Advances in Representation Theory of Algebras (ATRA VIII), Toruń, Poland (Sept 2022)*[Invited Speaker].
18. “Distribution of Schur representations of finite dimensional algebras”, Non-commutative algebras, representation theory and special functions, CRM, Montreal (May 2022)*[Invited Speaker].
19. “Distribution of bricks over tame algebras (Brick discrete vs. Brick continuous)”, Representations of Algebras and Related Combinatorics, Special Session of AMS meeting (March 2022).
20. “Classification of minimal τ -tilting infinite biserial algebras”, Homological Methods in Representation Theory (Conference in honor of Lidia Angeleri Hügel), Germany (Oct 2021).
21. “A modern analogue of the 2nd Brauer-Thrall conjecture”, Combinatorics and Representation Theory in Lausanne, Switzerland (Sept 2021).
22. “Moduli spaces of quiver representations and an open conjecture”, Mathematical Congress of Americas– Special session on Moduli Spaces in Algebraic Geometry and Applications (July 2021).
23. “Distribution of bricks over minimal representation infinite algebras”, Advances in Representation Theory of Algebras – ATRA VII (May 2021)*[Invited Speaker].
24. “Minimal τ -tilting infinite algebras and new conjectures”, 4th International Colloquium on representation of algebras and its applications, Bogota, Colombia (Nov 2020)*[Invited Speaker].
25. “A Categorification of biclosed sets of strings”, Categorifications in Representation Theory, Leicester, England (Sep 2020).

Seminars Talks (only after 2023).....

- 2026 : LACIM (May), Waterloo (May), Sherbrooke (March), Regina (March).
 2025 : Queen’s (Sept.), Carleton-Ottawa (Jan.), LACIM (Jan.), Sherbrooke (Jan.).
 2024 : Nantong (July).
 2023 : Tokyo-Nagoya (Dec.), OIST (Dec.), Bielefeld (Oct.), Verona (May), York (Jan.).

Organizing Duties:

Thematic Program Organization.....

- (Leading organizer) of the 7-week Thematic Program “*TDA PARTI - Topological Data Analysis, Persistence And Representation Theory Intertwined*”, Okinawa Institute of Science and Technology (OIST), Japan, June 23-August 9, 2025.
An interdisciplinary thematic program with over 70 in-person participants, with 5 Minicourses, 13 Expository presentations and 3 Colloquium talks, many discussion sessions, and a Symposium.

Conferences & Workshops Organization:.....

- (Leading organizer) of J-PEAKS Workshop “*IMPRINT: Interface of Multiparameter Persistence and Representation Theory Interleaved with New Topics*”, Okinawa, Japan, Feb. 1-5, 2027.
- Co-organizer of “*Spectrums in Representation Theory of Algebras and Related Topics*”, Osaka Central Advanced Mathematical Institute, Japan, Dec. 16–19, 2025.
- (Leading organizer) of “*Workshop on bricks and endofinite representations*”, Bielefeld, Germany, March. 3-5, 2025.
- (Leading organizer) of “*Autumn Research School & Conference: New Developments in Representation Theory of Algebras*”, Okinawa, Japan, Nov. 18-30, 2024.
- (Leading organizer) of “*Spring Workshop on Representation Theory of Algebras*”, Okinawa, Japan, April 22-26, 2024.

- (Local co-organizer) of the conference “*Advances in Representation Theory of Algebras IX*”, Kingston, Canada, June 12-16, 2023.
- (Leading organizer) of the scientific session on “*Representation theory of algebras*” at Canadian Mathematical Society, Toronto, Canada, Dec. 3-5, 2022.
- Co-organizer of the conference “*Representation Theory and Geometry*”, Feb. 14-16, 2022 [Online].

Research Schools Organization:.....

- (Leading organizer) of the CIRM Research School on “*GRAPE: Geometric Representation theory in multiparameter PERSISTENCE*”, CIRM Research School, Luminy, France, June 7-11, 2027.
- Co-organizer of “*Winter School on Large Modules and Endofiniteness*”, Stuttgart, Germany, Feb. 24-28, 2025.
- (Leading organizer) of the ISM Discovery School on “*Applications of Representation Theory in Topological Data Analysis & Geometric Invariant Theory*”, Montréal, Canada, June 3-7, 2024.
- Co-organizer of the ISM Discovery School on “*Mutations: from cluster algebras to representation theory*”, Montréal, Canada, July 4-8, 2022.

Working Groups & Seminars Organization:.....

- Founder and organizer of “Reading group on Machine Learning”, Queen’s University, 2022-23.
- Organizer of the “Algebra and Geometry Seminar” at Queen’s University, Canada, 2021-23.
- Co-founder and co-organizer of “LaCIM Representation Theory Working Group”, UQAM, 2015-19.
- Co-organizer of “Algebra Reading Group”, University of New Brunswick, 2014-15.

Teaching Experience (only at the university level)

Advanced Mini-courses (at different institutes):

The target audience for these advanced topics courses was other researchers in Representation Theory of Algebras, Algebraic Combinatorics, and related domains.

- *On second brick-Brauer-Thrall conjecture and related problems* [Mini-course, Remotely]
Topics course, consisting of three lectures (August 20-22, 2024), remotely hosted by IPM - Institute for Research in Fundamental Sciences, Isfahan, Iran.
- *On Combinatorics of gentle algebras* [Mini-course, in-person]
Topics course, consisting of six lectures (February 2024), at Institute of Science and Technology (OIST), Japan.
- *On minimal (τ -)tilting infinite algebras* [Mini-course, Remotely]
Topics course, consisting of six lectures (June 14-16, 2021), coordinated with Dr. Charles Paquette, remotely hosted by IPM - Institute for Research in Fundamental Sciences, Isfahan, Iran.

Undergraduate courses (at Queen’s University, Canada):

- *Linear Algebra II* [100 students– In-person], Winter 2023.
- *Group Theory* [40 students– In-person], Fall 2022.
- *Introduction to Linear Algebra* [150 students– In-person], Winter 2022.
- *Vector Calculus* [135 students– Hybrid], Fall 2021.
- *Differential and Integral Calculus* [one of the 3 coordinators, 1,300 students– Remotely], Winter 2020.
- *Vector Analysis* [80 students– Remotely], Fall 2020.

Teaching Assistance (at University of New Brunswick, Canada):

- Calculus for Management Science (Winter 2015), Introduction to Calculus II (Fall 2015), Introduction to Calculus I (Fall & Winter 2014), Multivariable Calculus (Winter 2013).

Services

- Reviewer and Referee in several journals of Mathematics (2020 - now), including Advances in Mathematics, Journal of Algebra, Journal of Algebra and Representation Theory, Journal of Algebra, Number Theory and Applications, Bulletin of London Mathematical Society, etc.
- Member of the advisory committee for graduate course arrangement and minicourses at Okinawa Institute of Science and Technology (June 2025 - Now)
- Committee member of racially visible minorities in Kingston, Canada (Sept 2022 - June 2023).
- Vice-President of Postdoctoral Scholars (Unit 2 of PSAC) at Queen's University, Kingston, ON, Canada (August 2022- March 2023).
- Reader in the Major Admission Awards Committee at Queen's University, Kingston, ON, Canada (2022 and 2023).
- Representative of Postdoctoral Scholars in the Postdoctoral Advisory Committee, Queen's University, Kingston, ON, Canada (July 2022 - June 23).
- Committee Member of Studc – Canadian Mathematical Society Student Committee (Sept. 2017 - April 2020).
- Representative of Graduate Students in Teaching Excellence and Policy Committee, University of New Brunswick, Fredericton, NB, Canada (April 2014 - April 2015).

References (In alphabetical order)

* For Research Capacity:

- Prof. Thomas Brüstle (Université de Sherbrooke, Canada)
Email address: thomas.brustle@usherbrooke.ca
- Prof. Osamu Iyama (University of Tokyo, Japan)
Email address: iyama@ms.u-tokyo.ac.jp
- Prof. Charles Paquette (Royal Military College, Canada)
Email address: charles.paquette@rmc.ca
- Prof. Hugh Thomas (Université du Québec à Montréal, Canada)
Email address: hugh.ross.thomas@gmail.com

* For Teaching & Professional Capacity:

- Dr. Ping Li (Queen's University, Canada)
Email address: pingli@queensu.ca
- Prof. Ram Murty (Queen's University, Canada)
Email address: murty@queensu.ca